

Binary Search Trees (Assignment Solutions)

Question 1 :

```
int rangeSumBST(TreeNode* root, int low, int high) {  
    if(root->val >=low && root->val <= high){  
        return root->val  
        + rangeSumBST(root->left,low,high)  
        + rangeSumBST(root->right,low,high);  
    }  
    if(root->val < low) {  
        return rangeSumBST(root->right,low,high);  
    }  
  
    return rangeSumBST(root->left,low,high) ;  
}
```

Question 2 :

```
void maxDiffUtil(struct Node *ptr, int k, int &min_diff,  
                int &min_diff_key) {  
  
    if (ptr == NULL)  
        return ;  
  
    if (ptr->key == k)  
    {  
        min_diff_key = 0;  
        return;  
    }  
  
    if (min_diff > abs(ptr->key - k))  
    {  
        min_diff = abs(ptr->key - k);  
        min_diff_key = ptr->key;  
    }  
  
    if (k < ptr->key)  
        maxDiffUtil(ptr->left, k, min_diff, min_diff_key);
```

```

        else
            maxDiffUtil(ptr->right, k, min_diff, min_diff_key);
    }

int maxDiff(Node *root, int k) {
    int min_diff = INT_MAX, min_diff_key = -1;
    maxDiffUtil(root, k, min_diff, min_diff_key);
    return min_diff_key;
}

```

Question 3 :

```

int search(TreeNode* root, int& K) {
    if (root == NULL) {
        return -1;
    }

    int leftVal = search(root->left, K);
    if(leftVal != -1) {
        return leftVal;
    }

    K--;
    if(K == 0) {
        return root->val;
    }

    return search(root->right, K);
}

int kthSmallest(TreeNode* root, int k) {
    return search(root, k);
}

```

Question 4 :

```

bool isPairPresent(Node* root, Node* temp, int target)
{

```

```
        if (temp == NULL)
            return false;
        return search(root, temp, target - temp->data)
            || isPairPresent(root, temp->left, target)
            || isPairPresent(root, temp->right, target);
    }
    bool search(Node* root, Node* temp, int k)
    {
        if (root == NULL)
            return false;
        Node* c = root;
        bool flag = false;
        while (c != NULL && flag != true) {
            if (c->data == k && temp != c) {
                flag = true;
                cout << "Pair Found: " << c->data << " + "
                    << temp->data;
                return true;
            }
            else if (k < c->data)
                c = c->left;
            else
                c = c->right;
        }
        return false;
    }
}
```

Question 5 :

```
class nodeValue{
public:
    int min;
    int max;
    int sum;
    nodeValue(int min,int max,int sum)
    {
        this->min=min;
```

```

        this->max=max;

        this->sum=sum;

    }

};

class Solution {
public:
    int ans=0;

    int maxSumBST(TreeNode* root) {
        helper(root);

        return ans;
    }

    nodeValue helper(TreeNode* root) {
        if(root==NULL)
            return nodeValue(INT_MAX, INT_MIN, 0);
        nodeValue l=helper(root->left);
        nodeValue r=helper(root->right);
        if(root->val>l.max && root->val<r.min)
        {
            ans=max(ans, root->val+l.sum+r.sum);
            return
nodeValue(min(root->val, l.min), max(root->val, r.max), l.sum+root->val+r.sum);
        }
        return nodeValue(INT_MIN, INT_MAX, max(l.sum, r.sum));
    }
};

```